GeoVac: A Field Guide
From a Packing Puzzle to Periods of the Spectral Triple

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Abstract

The Geometric Vacuum (GeoVac) framework began as a packing puzzle that produced something resembling the periodic table. Over the course of the investigation, it acquired a precise structural identity: a discrete spectral triple that is an instance of the Marcolli–van Suijlekom 2014 graph-network construction at the data level and extends it via a forced-graph claim and a master Mellin engine, whose transcendentals classify as cyclotomic mixed-Tate periods at level ≤ 4, and which separates — through theorems, not preference — what the underlying geometry forces from what empirical measurement supplies. This field guide is the project's identity statement. It tells the arc from origin to current frontier without recapitulating the technical machinery, and points each kind of reader toward the right entry into the corpus. As the project's *grand synthesis* it now sits atop a complete layer of six audience-targeted group syntheses—one per branch (Groups 1–6), each digesting its branch in a few pages—and composes them into a single front-to-back reading that routes each reader to the right branch digest. The one-sentence reading: *GeoVac is a discrete spectral triple whose periods classify the projective content of physics, sharply separating what geometry forces from what measurement supplies.*

Contents

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1 How This Started

A short note on the path that led to the framework, told in the first person. The reader can skip it without missing technical content; the arc proper begins in §2.

In high school I looked at the periodic table and wondered why it had that shape — in particular, why the rows didn't grow uniformly but in a stepped pattern. I sketched what I thought might be a more natural picture: a circular layout with the first row at the centre and successive rows fanning outward. I had no mathematics to back it up; it was a visual hunch.

In college I learned summation notation and the discipline of inductive proof, and I felt — for the first time — that I had permission to do real mathematics. I went back to the periodic table. The row lengths grew by a recurring +4 between successive pairs. I built a nested summation that captured the pattern and arrived at $2n^2$. I looked up the number. It was the degeneracy of the Schrödinger principal quantum level. I felt something between satisfaction and disappointment: the pattern was already known — to Schrödinger, who had derived it by solving a differential equation on $\mathbb{R}^3$. My combinatorial route to the same number was a curiosity, not a discovery. I set the question down.

Years later, listening to a guest on Sean Carroll's podcast, I heard the suggestion that perhaps, instead of quantising gravity, one should try to quantise spacetime — a discrete substrate as foundation, not as a tool for approximating something continuous. The framing revived the old question. This time I posed it in polar coordinates and treated Planck's constant as a unit volume to be packed. I wrote down a small set of packing axioms. The construction, executed honestly, produced circles whose node counts reproduced the periodic table.

> *This struck me as deeply important.*

I wrote a first paper that I would now describe as a book report: a narrative explanation of how the packing maps to the periodic table, without any of the spectral or geometric machinery that came later. That paper became Paper 0 [14] in its mature form. I then set out to see how the nodes might relate to quantum-number counting. Everything in the rest of this field guide — the Fock 1935 identification, the natural-geometry hierarchy, the spectral-triple rotation, the periods classification, the forced-versus-free seam — is what happened next.

What follows is the technical arc.

2 The Packing Puzzle

The framework's origin is a geometric construction: pack Planck's constant. Treat $\hbar$ as a unit volume to be filled, and ask which node arrangements complete the packing without gaps. In two dimensions the node counts give (l, m): the integer pair indexing azimuthal and magnetic quantum numbers, before any quantum-mechanical input. In three dimensions, an inductive argument over periodic-table row lengths fixes the principal and spin quantum numbers (n, s) [14]. The resulting graph has nodes labeled by quantum-number tuples (n, l, m, s) and edges encoding spectral selection rules.

This graph is conformally equivalent to a continuous operator on the unit three-sphere (the Laplace–Beltrami operator; see §3), whose eigenvalues are

$$\lambda_n = -(n^2 - 1), \qquad n \in \mathbb{N}_{\geq 1}, \tag{1}$$

with multiplicity n^2 — the spectrum of the hydrogen atom, to within an overall scale. The shape of the periodic table fell out of the packing.

This was the seed: a combinatorial object that, when interrogated, returned the same numerology as observed atomic structure. Whether that was a coincidence or a clue was the question the rest of the project tried to answer.

3 Fock 1935: Discrete and Continuous Are the Same Thing

In 1935 Vladimir Fock observed [1] that the hydrogen atom on $\mathbb{R}^3$ has a hidden $SO(4)$ symmetry, and that the stereographic projection $\mathbb{R}^3 \hookrightarrow S^3$ converts the Schrödinger equation into an angular eigenvalue problem on the unit three-sphere. What hydrogen really is, from this point of view, is the Laplace–Beltrami operator on the unit S^3; the $1/r$ Coulomb potential is the coordinate distortion produced by the projection (specifically, the chordal-distance identity that takes the round metric to the stereographic one).

The decisive observation for GeoVac is this: the graph Laplacian on the packed lattice of §2 is conformally equivalent to the continuous S^3 Laplace–Beltrami operator, in a precise sense that can be checked symbolically. Paper 7 of the GeoVac corpus [16] provides 18 symbolic proofs of this equivalence (`tests/test_fock_projection.py` and `tests/test_fock_laplacian.py`). The universal scaling constant relating the two is $\kappa = -1/16$; its value *coincides* with a conformal-geometry quantity — a numerical observation, not a derivation (Paper 0 [14], Paper 7 [16]).

This is the framework's load-bearing identification:

> *The discrete Fock graph and the continuous unit-S^3 Laplace–Beltrami operator are not two models of the hydrogen atom. They are the same object in two coordinates.*

The implication is operational: where the continuous formulation requires integration over $\mathbb{R}^3$ with a $1/r$ singularity, the discrete formulation requires a sparse matrix eigenvalue problem with integer entries. Same physics, different cost.

4 Natural Geometry: a Hierarchy

Section 3's identification works because S^3 is the natural geometry for the hydrogen atom: it is the coordinate system in which the Schrödinger equation separates fully into angular variables. For other systems — molecules, multi-electron atoms, polyatomics — the natural geometry shifts. The framework's chemistry investigation (2024–2026, Phase 2 of the project) systematically catalogued the natural geometries:

Level	System	Natural geometry	Best accuracy	Paper
1	H (1-center, 1e)	S^3 (Fock)	$< 0.1\%$	7
2	H_2^+ (2-center, 1e)	Prolate spheroid	0.0002%	11
3	He (1-center, 2e)	Hyperspherical	0.004% (cusp)	13
4	H_2 (2-center, 2e)	Mol-frame hyperspherical	96.0% D_e	15
5	LiH (composed)	Composed (Level 3 $\otimes$ Level 4)	5.3% R_{eq}	17

The choice of geometry is determined by where separation of variables occurs; in each case the discrete graph Laplacian on the natural geometry reproduces the continuous Schrödinger problem with the same conformal-equivalence identification as in §3.

Phase 2 reached known structural ceilings. Heavy molecules ($Z \geq 20$ or strong multi-determinant correlation) escape the framework's accuracy targets through identifiable channels (the "multi-focal-composition wall" — the project's name for a structural obstruction established by multiple independent negative results — documented across six independent diagnostic investigations). The framework's most natural quantum-simulation application — Pauli-operator counts exactly linear in the qubit count Q at fixed basis ($N_{\text{Pauli}} = 27.90 \times Q$) for the core–valence "composed" encoding, with a sub-quadratic $O(Q^{1.77})$ 1-norm, benchmarked across a 37-system library [17] — shipped at the end of Phase 2. The absolute Pauli multipliers are convention-sensitive (near parity with minimal-basis STO-3G, 76× sparser than cc-pVDZ); the load-bearing advantage is the scaling, not any single market-test ratio.

But a different question began to dominate the conversation around the project, and that question is where the rest of this guide goes.

5 The Rotation: from Solver to Spectral Triple

Phase 3 was planned as quantum simulation. Instead, the project rotated to a different question: *what is this thing we have been working with?*

The graph Laplacian on S^3, the conformal equivalence with the continuous problem, the natural-geometry hierarchy that produces discrete substrates for every electron configuration — these are collectively the data of a Riemannian spectral geometry. Made precise, that means an algebra of functions, a Hilbert space of states, and a Dirac-type operator, all assembled into a structure that mathematicians call a spectral triple [3].

The precise structural claim:

> *GeoVac is a discrete spectral triple that is an* instance *of the Marcolli–van Suijlekom 2014 gauge-network construction [5] at the data level (refined by the Perez-Sanchez clarification [6, 7] that the continuum limit is Yang–Mills without Higgs in general) and an* extension *of it at the structural level via the forced-graph claim (Bertrand $\times$ Fock projection), the master Mellin engine, and the cosmic-Galois layer U^* (Paper 56).*

This claim was checked, not assumed — and then re-checked. A five-lemma Gromov–Hausdorff convergence theorem on truncated Camporesi–Higuchi spectral triples [22] (Paper 38) gives the quantitative statement, in van Suijlekom's state-space Gromov–Hausdorff framework (adjacent to, but distinct from, the Latrémolière propinquity [8]), with rate

$$\Lambda_{n_{\max}} \leq C_3 \cdot \gamma_{n_{\max}}, \qquad \gamma_{n_{\max}} \sim \frac{4}{\pi} \frac{\log n_{\max}}{n_{\max}} \xrightarrow{n_{\max}\to\infty} 0, \tag{2}$$

with the constant $4/\pi$ universal across compact simple Lie groups (rigorous at rank 1, numerically pinned at higher ranks; the rank-uniform proof is a named gap) [23]. Paper 38's theorem is unconditional in van Suijlekom's state-space framework, via a translation-seminorm metrization (the kernel condition is proved on the truthful substrate; the dual direction uses an exact-fit spinor lifted state, and both almost-inverse defects are Fejér smoothings at the same moment). See Paper 38 § 1 and the claims register, `docs/claims_register.md`. The rate constant $4/\pi$ is the substrate's $\mathrm{Vol}(S^2)/\pi^2$ signature — the Hopf-base measure entering through a single geometric mechanism, classified in §6 below.

Following Paper 38, the project produced a math.OA arc extending the framework along the Lorentzian axis (Papers 42–49) [24, 25, 26, 27, 28, 29, 30, 31]. The most instructive outcome of that axis is *negative*: the natural device for a “Lorentzian propinquity” — compressing the Krein Dirac to the Krein-positive subspace — annihilates the spatial Dirac, so the compressed Lipschitz seminorm vanishes identically (Paper 45's degeneracy theorem [27], with a bit-exact frozen falsifier). An earlier convergence claim through that device, together with the *metric-level* strong-form and bridge convergence claims built on it (Papers 46, 48, 49), is withdrawn; the papers of record carry Status notes. What survives: the operator-system substrate (Paper 44), the four-witness modular-Hamiltonian results (Papers 42/43), the norm-resolvent de-compactification arrow (Paper 47), and the cocycle-deficit algebra of Paper 49 (whose operator-system Lorentzian pre-length-space bridge stands at the algebra level). The repaired Lorentzian target — a Toeplitz temporal multiplier algebra and a non-compressing Krein-compatible seminorm — is stated as an open problem in Paper 45.

Why this rotation mattered. The shift from "how accurately can we solve quantum chemistry?" to "what is GeoVac?" produced the right kind of answers. Chemistry accuracy is a sliding scale — always tighter at more cost. But the spectral-triple structural reading is binary: either GeoVac IS the data of a spectral triple, or it isn't. Theorem-grade closure is available either way. The framework's identity is now in the first category.

6 Periods: Where Transcendentals Come From

A spectral triple has a heat kernel, and the heat kernel produces transcendentals. In GeoVac, those transcendentals are not random. They sit in named classes, with named mechanisms producing them.

The graph itself is π-free. At what the project calls *Layer 1* — the bare combinatorial graph and its integer eigenvalue spectrum — there are no transcendentals. Matrix elements are rationals (or, at most, algebraic over $\mathbb{Q}$). This has been checked across the corpus: verified explicitly on the unit S^3 Coulomb spectrum (Paper 7), on the Hopf bundle gauge structure (Paper 25), on the Bargmann–Segal S^5 Hardy lattice (Paper 24), and on the 37 molecular and atomic species in the qubit Hamiltonian library (Paper 14).

Transcendentals enter at projection. Layer 1 by itself does not produce physical observables. Observables come from *Layer 2:* projections of the discrete substrate onto continuous manifolds, integration regions, or measurement-coordinate choices. Each projection introduces a controlled transcendental through a controlled mechanism.

The project's full classification fits in three named mechanisms, collectively the *master Mellin engine* [18, 21, 32]:

Slot k	Mechanism	Period-ring home	Source
$k = 0$	M1 (Hopf-base measure)	$\mathbb{Q}[\pi, \pi^{-1}]$	Paper 55 §3
$k = 2$	M2 (Seeley–DeWitt)	$\bigoplus_k \pi^{2k} \cdot \mathbb{Q}$	Paper 55 §4
$k = 1$	M3 (vertex parity / Hurwitz / Dirichlet L)	$\mathcal{MT}(\mathbb{Z}[i, 1/2])$ at level ≤ 4	Paper 55 §5

The three rows are the three sub-cases of a single Mellin transform

$$\mathcal{M}\Big[\mathrm{Tr}(D^k \cdot e^{-tD^2})\Big]\,(s) \tag{3}$$

on the discrete Camporesi–Higuchi S^3 spectral triple, with operator-order index $k \in \{0, 1, 2\}$ selecting which mechanism produces which kind of transcendental.

The classification is complete. Paper 32 § VIII [21] contains a *case-exhaustion theorem*: every appearance of π in any closed-form GeoVac spectral observable arises from one of M1, M2, or M3. No fourth mechanism has been found despite specific searches. Paper 55 [32] promotes this from a partition theorem to a period-theoretic classification: every transcendental observed in the corpus sits in cyclotomic mixed-Tate periods at level ≤ 4 over $\mathbb{Z}[i, 1/2]$ (Kontsevich–Zagier [9], Brown [10], Deligne [11], Glanois [12], Fathizadeh–Marcolli [13] are the relevant published classifications; each piece of GeoVac's data is placed inside one of their named period rings).

What was a zoo became a classified program. Early in its development the project produced transcendentals through ad-hoc calculations and was uncertain whether different appearances of π or $\zeta(2)$ were "the same" or "different." Today every transcendental observation has a named motivic home, indexed by which slot of (3) produced it. The classification makes the operational question — "what kind of transcendental can appear from *this* calculation?" — algorithmic.

A note on periods. A period in the sense of Kontsevich–Zagier 2001 [9] is a complex number whose real and imaginary parts can be expressed as integrals of algebraic functions over domains defined by polynomial inequalities with rational coefficients. π, $\log 2$, $\zeta(n)$ for $n \geq 2$, Catalan's constant $G = \beta(2)$, the Dirichlet $\beta(s)$ values at integer s — all are periods. The cyclotomic mixed-Tate sub-class at level ≤ 4 over $\mathbb{Z}[i, 1/2]$ is, conjecturally, a small and well-understood corner of the full period algebra. GeoVac sits in this corner.

7 The Forced/Free Seam

The master Mellin engine of §6 classifies what GeoVac *generates*. An equally important question is what GeoVac *does not* generate — in particular, the empirical constants of physics (the fine-structure constant α, the Yukawa couplings, the generation count N_{gen}, the cosmological constant) that must enter as external inputs. The framework's most operationally useful structural discipline is the clean separation between what geometry forces and what measurement supplies.

What geometry forces (the structural skeleton). The following are all theorem-grade forced by the framework's construction, independent of any empirical input:

- The outer manifold is S^3. Forced by Bertrand's theorem [2] (only $1/r$ and harmonic-oscillator central potentials produce closed bound orbits, of which $1/r$ is the Coulomb case) plus the $SO(4)$ hidden symmetry of the Coulomb problem (Paper 7).
- The Hopf bundle structure $S^3 \to S^2$ is forced by the maximal-torus reduction of $SU(2) = S^3$ (Paper 25) [19].
- The Camporesi–Higuchi Dirac sector is forced by the parallelizability of S^3 (Paper 32) [21].
- The Wilson $SU(2)$ gauge structure is forced by the non-abelian generalization of the abelian Hopf structure (Paper 30) [20].
- The Standard Model gauge group $U(1) \times SU(2) \times SU(3)$ truncated at $n \leq 3$ in the complex-Hopf tower $S^{2n-1} \to SU(n)$ is forced by Bertrand $\times$ Hopf-tower structural saturation (Paper 32 § VIII).
- The dimension of the inner Dirac moduli space is $\dim \mathcal{M}(D_F) = 128$ per generation, 8 in the diagonal slice. Bertrand-shaped Forced-Count Theorem (Paper 32 § VIII, June 2026).

What measurement supplies (calibration data). The following are all empirical inputs to the framework, falling outside any structural derivation it offers:

- The fine-structure constant $\alpha^{-1} \approx 137.036$. Three independent spectral homes for the framework's combination $K = \pi(B + F - \Delta)$ have been established (B as Casimir trace, $F = \zeta(2)$ as Fock-Dirichlet, $\Delta = 1/40$ as Dirac mode degeneracy), but the combination rule itself remains a numerical observation, not derived. Paper 2 is in the Observations folder [15]; twelve mechanisms have been eliminated for a single-principle derivation of K.

- The Yukawa coupling values. A 162-cell PSLQ sweep against low-coefficient pure-Tate periods returned zero hits. The Yukawa Dirichlet ring lives in Paper 18 § IV.6's sixth tier (parameter-tied Dirichlet, categorically disjoint from M1/M2/M3).
- The generation count $N_{\text{gen}} = 3$. The Hopf-tower-to-representation extension was ruled out as a shortcut for forcing N_{gen}: the "3 algebra factors" and the "3 generations" are different 3s in the standard SM representation. A deep structural obstruction; no known line of attack.
- The cosmological constant. The framework's CC formula $\Lambda_{\text{cc}} = 6\phi(2)/\phi(1) \cdot \Lambda^2$ inherits the standard $\sim 10^{120}$ fine-tuning problem (Paper 51 G7).

The seam is itself a theorem. What makes the forced/free separation operationally useful, rather than rhetorical, is that the boundary between the two categories is itself controlled by named theorems:

- The Forced-Count Theorem (Paper 32 § VIII): the gauge structure of GeoVac is forced.
- The Yukawa Non-Selection Theorem (Paper 32 § VIII): the framework's almost-commutative extension admits a Higgs structure but does not autonomously select the Yukawa matrix.
- The η-Trivialization Theorem and AC-Factorization Theorem (Paper 18 § IV.6): the combined heat trace on the AC extension factorises cleanly into an outer pure-Tate factor and an inner Dirichlet factor in the Yukawa variables, with no mixing.

These three theorems collectively pin the seam: values are free, relations are forced. This is the project's most operationally useful structural reading.

Theorem-bounding the seam (June 2026). The forced/free seam discipline above was sharpened into a set of named structural theorems in June 2026. Paper 57 catalogues sixty named entries across nine domains (35 forced, 1 admitted-not-forced, 24 calibration) and tests a "packing-reachability" discriminator (P5); its 98.3% agreement with the forced/free labelling is an internal-consistency check, not an independent discriminator — the packing-reachable tag is the forced/free label by construction (Paper 57 §6.1), and the genuine discriminator signal is the two-family (multi-focal composition vs. inner-factor input-data) decomposition. Paper 32 § VIII consolidates eight theorem-grade non-selection results (Forced-Count, H1 Yukawa, N_{gen}, KO-dimension, single-cutoff spectral action, spatial-composition wall, no-single-mechanism-K, and the cutoff function as external test-function data) across four sectors: inner-factor structural skeleton, multi-focal composition, α-combination, and gravity. The seam's individual entries are thus theorem-bound; but Paper 57 remains an observations paper — the meta-theorem that would decide forcedness *without* first locating a witness derivation is still open, with a partial Bohr-site answer (Paper 57's Bohr-site dichotomy: forced = site property, calibration = valuation, admitted = site degeneracy).

8 What's in the Corpus

The GeoVac corpus is 50+ papers organised into six audience-targeted groups (`papers/group1_operator_alge` etc.) plus an archive and a synthesis folder. Each group is described paper-by-paper in the repository's `papers/INDEX.md`. This section gives one-line summaries.

- **Group 1 — Operator algebras / NCG** (16 papers). Math.OA arc. Headline: unconditional SU(2) state-space Gromov–Hausdorff convergence with explicit $4/\pi$ rate (Paper 38, translation-seminorm metrization); the Lorentzian degeneracy theorem (Paper 45) closing the natural Lorentzian route and stating the repaired target; four-witness modular-Hamiltonian identification (Papers 42/43). Papers 46–49 carry Status notes for descoped metric-level claims.

- **Group 2 — Quantum chemistry** (9 papers). Natural-geometry hierarchy levels 1–5. Headline: structural accuracy ceilings characterized via the multi-focal-composition wall; Paper 8–9 and FCI-M are guardrail papers documenting proven negative results.

- **Group 3 — Foundations** (11 papers). Headline: Paper 0 (packing), Paper 7 (Fock-projection conformal equivalence), Paper 22 (angular sparsity), Paper 24 (Bargmann–Segal S^5 lattice), Paper 31 (universal/Coulomb partition), Paper 55 (periods classification).

- **Group 4 — Quantum computing** (4 papers). Qubit-encoding resource estimates. Headline: exactly linear composed Pauli scaling in Q at fixed basis ($N_{\text{Pauli}} = 27.90 \times Q$) and a sub-quadratic $O(Q^{1.77})$ 1-norm across a 37-system library (Papers 14/20); absolute Pauli multipliers are convention-sensitive (near parity with STO-3G, 76× vs cc-pVDZ).

- **Group 5 — QED / gauge / SM** (8 papers). Headline: hydrogen Lamb shift at -0.534% one-loop closure (Paper 36); SU(2) Wilson lattice gauge structure with maximal-torus reduction to the U(1) of Paper 25 (Paper 30); QED on S^3 with five theorems and the two-loop constant S_{min} identified in closed form (Paper 28).

- **Group 6 — Precision observations** (4 papers). Headline: five Roothaan autopsies decomposing observables into projection-chain contributions (Paper 34); time as projection (Paper 35); universal S_B scaling on sparse lattices (Paper 27).

The `papers/archive/` folder contains historical and superseded drafts. The `papers/synthesis/` folder contains this field guide together with a complete layer of *six* group-level syntheses—one per audience branch (Groups 1–6), each digesting its branch in four to six pages. This field guide is the grand synthesis that composes them; a reader who wants a single branch should read that branch's group synthesis directly.

9 What's Open

The project is honest about scope. The following are named open questions, in rough order of how much external infrastructure each would require:

- **The α-derivation**. Paper 2's combination rule $K = \pi(B + F - \Delta)$ matches the empirical α^{-1} at 8.8×10^{-8} relative precision, but is classified as a numerical observation, not a derivation. Twelve mechanisms eliminated for a single-principle derivation; the project's standing internal reading is that α is a projection constant, not a derivable number.

- **The Yukawa values**. A dedicated PSLQ search ruled out the Yukawas as low-coefficient pure-Tate periods. They are Class 1 calibration data and live in Paper 18 § IV.6's sixth tier. Whether they admit a deeper non-period derivation is open and falls outside the master Mellin engine's classification scope.

- $N_{\text{gen}} = 3$. A deep structural obstruction; no line of attack is known.

- **The cosmological constant problem**. Standard fine-tuning $\sim 10^{120}$ inherited from the Connes–Chamseddine spectral action expansion (Paper 51 G7). No GeoVac-specific resolution proposed.

- **The chemistry correlation wall (W1e)**. Multi-determinant FCI correlation in second-row and heavier diatomic binding is below the framework's structural accuracy ceiling. Six independent diagnostic investigations identified the obstruction as multi-focal-composition rather than a missing single mechanism.

- **The M2/M3 motivic Galois unification (Q5')**. Both M2 and M3 engage the cyclotomic field $\mathbb{Q}(i) = \mathbb{Q}(\zeta_4)$, but via structurally independent mechanisms. Whether a single Tannakian symmetry unifies them is an open mathematical research project (Paper 55 § 7.6). A deep frontier.

- **Deligne–Milne pro-finite reconstruction**. Paper 56 closes the cosmic-Galois inclusion $\Phi : U^{*(n_{\max})} \hookrightarrow \mathrm{Aut}^{\otimes}(\omega)$ at finite cutoff via the Tannakian dual of an explicit Hopf algebra. The converse equality $\mathrm{Aut}^{\otimes}(\omega) = \lim_{\leftarrow} U^{*(n_{\max})}$ at the pro-finite limit (Deligne–Milne 1982 Theorem 2.11) is the named deep frontier; every intermediate step nonetheless closed readily at finite cutoff, suggesting the pro-finite assembly may prove similarly tractable once the right transition-map machinery is in place.

- **Non-commutative Mondino–Sämann (Q2')**. Paper 49 closes the strong-form Krein–MS bridge (Q1$'$) on the enlarged-substrate chirality-flipping generators. The non-commutative analog of Mondino–Sämann pre-length-spaces (a non-commutative Lorentzian GH program built from spectral-triple data rather than causal diamonds) is the remaining substantial NCG-research target.

- **Inhomogeneous extensions**. The full Fathizadeh–Marcolli generic Robertson–Walker spacetime with non-trivial scaling factor $a(t)$ requires either a continuum substrate or a time-dependent Camporesi–Higuchi spectral triple, outside the discrete-substrate principle. The closest GeoVac analog is the discrete-warped-substrate program of Paper 51 §G4-3 (named follow-on).

- **Full Standard Model spectral triple**. The Connes Standard Model on S^3 alone was closed in May 2026 — positive, but structurally thin; the genuine cross-manifold construction $S^3 \otimes \mathrm{Hardy}(S^5)$ is blocked at the Coulomb-versus-harmonic-oscillator asymmetry of Paper 24 § V at layer 4 (modular Hamiltonian of the wedge KMS state). Resolution requires NCG-framework extension.

What is *not* open — but is sometimes asked — is whether GeoVac is a new foundation for quantum mechanics. It is not. GeoVac is a discretization framework whose discrete substrate is conformally equivalent to known continuous quantum mechanics at every level where the equivalence has been tested. The math supports both readings; the interpretive question is left to the reader.

10 Reader's Map

This is a multi-paper corpus. Here is where each kind of reader should start. Each audience now has a one-document digest—its *group synthesis*—which is the recommended first stop before the individual papers: Group 1 (operator algebras), Group 2 (quantum chemistry), Group 3 (foundations), Group 4 (quantum computing), Group 5 (QED / gauge / gravity), and Group 6 (precision observations). The per-paper entries below are the second step.

- **The mathematician (NCG, math.OA, spectral triples)**. Start with the three-page front-door note (`docs/outreach/note_n1_su2_truncations.pdf`), then Papers 38, 45, and 32. Paper 38 is the unconditional SU(2) state-space Gromov–Hausdorff convergence theorem (translation-seminorm metrization, compression/lifted-state proof); Paper 45 is the Lorentzian degeneracy theorem (its History remark records a withdrawn earlier convergence claim); Paper 32 is the explicit spectral triple construction with Connes axiom audit and the § VIII C-arc closure of eight theorem-grade non-selection results. Papers 42/43 close the four-witness modular-Hamiltonian identification; Paper 47's norm-resolvent arrow stands; Papers 46, 48, and 49 carry Status notes.

- **The mathematician (periods, motives, Tannakian)**. Read Papers 55 and 56. Paper 55 places GeoVac's periods in the cyclotomic mixed-Tate ring at level ≤ 4 with explicit Glanois descent on the vertex-parity Mellin slot. Paper 56 identifies the cosmic-Galois group $U^* = \mathbb{G}_a^{3N} \rtimes SL_2$ at finite cutoff via the Tannakian dual of an explicit Hopf algebra $\mathcal{H}_{\mathrm{GV}}$, with 5,864 bit-exact zero residuals across the Pro-System + Tannakian closure panel. Inclusion direction closed at finite cutoff; converse equality with the Deligne–Milne reconstruction at pro-finite limit is the named deep frontier. Open M2/M3 unification question (Q5') discussed in Paper 55 § 7.6 and Paper 56's open-questions section.
- **The physicist (atomic, molecular, optical)**. Read Papers 7, 22, and 14. Paper 7 is the Fock-projection conformal equivalence; Paper 22 is the angular sparsity theorem (universal across central potentials); Paper 14 is the quantum-computing resource benchmarks.
- **The physicist (high-energy, gauge, SM)**. Read Papers 30, 25, 28, and 36. Paper 30 is SU(2) Wilson on S^3; Paper 25 is the abelian-Hopf reduction; Paper 28 is QED on S^3; Paper 36 is the bound-state QED arc closing the Lamb shift at -0.534% one-loop.
- **The physicist (gravity)**. Read Paper 51. Spectral action approach, two-term exact on S^3; the $S_{\mathrm{BH}} = A/4$ replica derivation was closed by the replica-weight-harmless theorem.
- **The quantum computing practitioner**. Read Paper 14 and Paper 20. Pauli scaling, OpenFermion/Qiskit/PennyLane export, market test against published Gaussian baselines.
- **The chemist**. Read Papers 11, 13, 15, 17. The natural-geometry hierarchy, levels 2–5. Note the guardrail papers 8–9 and FCI-M before proposing any single-center or LCAO molecular encoding (these document proven negative results).

The CLAUDE.md document in the repository root is the project's internal coordination file; it indexes every paper, every sprint, every working hypothesis, and every documented failed approach. External readers do not need to read it but may find the "Failed approaches" table (§ 3) instructive for understanding the boundary between what works and what doesn't.

11 Conclusion

The one-sentence reading: *GeoVac is a discrete spectral triple whose periods classify the projective content of physics, sharply separating what geometry forces from what measurement supplies.*

The story is a journey from a packing puzzle through Fock 1935's S^3 identification, through the natural-geometry hierarchy and its chemistry investigation, to a structural reading as a discrete spectral triple that instances Marcolli–van Suijlekom 2014 at the data level and extends it via the forced-graph claim and master Mellin engine, with transcendentals classified as cyclotomic mixed-Tate periods at level ≤ 4, and with the forced-versus-free seam pinned by three named theorems.

The discipline is the separation: what geometry forces gets a name, a theorem, and a paper; what measurement supplies is recorded as calibration data and stays calibration data unless the framework can produce a derivation, which so far it cannot.

The project began as an amateur geometric puzzle and has acquired the status of a discrete spectral triple capable of closing published-open questions in the math.OA literature. That trajectory was AI-augmented at every step: implementation, sprint memos, literature collation, paper drafting, and the internal coordination machinery were all developed collaboratively with Anthropic's Claude, with PI direction and quality control.

The invitation: come read. The corpus is dense, but the entry points are now indexed. Section 10 above is the reader's map.

Acknowledgments. This work was developed in an AI-augmented agentic workflow with Anthropic's Claude. Implementation, internal sprint memos, paper drafting, and bibliographic collation were performed collaboratively; all theorems, design choices, and editorial decisions are the author's responsibility. The project has no institutional affiliation and is disseminated through DOI-stamped Zenodo releases.

References

[1] V. Fock, "Zur Theorie des Wasserstoffatoms," *Z. Physik* **98**, 145–154 (1935).

[2] J. Bertrand, "Théorème relatif au mouvement d'un point attiré vers un centre fixe," *C. R. Acad. Sci. Paris* **77**, 849–853 (1873).

[3] A. Connes, *Noncommutative Geometry*, Academic Press (1994).

[4] A. Connes and W. D. van Suijlekom, "Spectral truncations in noncommutative geometry and operator systems," *Commun. Math. Phys.* **383**, 2021–2067 (2021).

[5] M. Marcolli and W. D. van Suijlekom, "Gauge networks in noncommutative geometry," *J. Geom. Phys.* **75**, 71–91 (2014); preprint arXiv:1301.3480 (2013).

[6] C. I. Perez-Sanchez, "Bratteli networks and the Spectral Action on quivers," arXiv:2401.03705 (2024).

[7] C. I. Perez-Sanchez, "Comment on 'Gauge networks in noncommutative geometry'," arXiv:2508.17338 (2025).

[8] F. Latrémolière, "The dual Gromov-Hausdorff propinquity," *J. Math. Pures Appl.* **103**, 303–351 (2015).

[9] M. Kontsevich and D. Zagier, "Periods," in *Mathematics Unlimited — 2001 and Beyond*, Springer (2001), pp. 771–808.

[10] F. Brown, "Mixed Tate motives over $\mathbb{Z}$," *Ann. of Math.* (2) **175**, 949–976 (2012).

[11] P. Deligne, "Le groupe fondamental unipotent motivique de $\mathbb{G}_m - \mu_N$, pour $N = 2, 3, 4, 6$ ou 8," *Publ. Math. IHES* **112**, 101–141 (2010).

[12] C. Glanois, "Motivic unipotent fundamental groupoid of $\mathbb{G}_m \setminus \mu_N$ for $N = 2, 3, 4, 6, 8$ and Galois descents," arXiv:1411.4947 (2015); *J. Number Theory* **160**, 334–384 (2016).

[13] F. Fathizadeh and M. Marcolli, "Periods and motives in the spectral action of Robertson–Walker spacetimes," *Commun. Math. Phys.* **356**, 641–671 (2017); preprint arXiv:1611.01815 (2016).

[14] J. Loutey, "Paper 0: Geometric Packing and the Universal Constant $\kappa = -1/16$," GeoVac corpus (2026).

[15] J. Loutey, "Paper 2: $\alpha = K^{-1}$ as a spectral coincidence on the Fock-projected S^3," GeoVac corpus (2026).

[16] J. Loutey, "Paper 7: The Dimensionless Vacuum on S^3," GeoVac corpus (2026).

[17] J. Loutey, "Paper 14: Qubit encoding, Pauli scaling, and composed sparsity," GeoVac corpus (2026).

[18] J. Loutey, "Paper 18: Exchange constants and the master Mellin engine," GeoVac corpus (2026).

[19] J. Loutey, "Paper 25: GeoVac Hopf graphs as discrete lattice-gauge structure," GeoVac corpus (2026).

[20] J. Loutey, "Paper 30: SU(2) Wilson lattice gauge on $S^3 = SU(2)$," GeoVac corpus (2026).

[21] J. Loutey, "Paper 32: The GeoVac Spectral Triple," GeoVac corpus (2026).

[22] J. Loutey, "Paper 38: State-space Gromov–Hausdorff convergence of truncated Camporesi–Higuchi spectral triples," GeoVac corpus (2026); released to Zenodo 2026-05-07.

[23] J. Loutey, "Paper 40: State-space Gromov–Hausdorff convergence of spectral truncations of compact Lie groups with bi-invariant metric," GeoVac corpus (2026).

[24] J. Loutey, "Paper 42: Tomita–Takesaki modular structure and the four-witness Wick-rotation theorem," GeoVac corpus (2026).

[25] J. Loutey, "Paper 43: Lorentzian extension at finite cutoff via Krein-space at signature (3, 1)," GeoVac corpus (2026).

[26] J. Loutey, "Paper 44: Operator-system extension of the BBB Krein spectral triple," GeoVac corpus (2026).

[27] J. Loutey, "Paper 45: Lorentzian propinquity on truncated SU(2)×U(1)$_T$ Krein spectral triples: a degeneracy theorem," GeoVac corpus (2026).

[28] J. Loutey, "Paper 46: Strong-form Lorentzian propinquity on truncated SU(2)×U(1)$_T$ Krein spectral triples," GeoVac corpus (2026).

[29] J. Loutey, "Paper 47: Norm-resolvent convergence to the non-compact Lorentzian Krein spectral triple," GeoVac corpus (2026).

[30] J. Loutey, "Paper 48: Krein-pointed proper QMS and the bridge to Mondino–Sämann Lorentzian pre-length spaces," GeoVac corpus (2026).

[31] J. Loutey, "Paper 49: OSLPLS strong-form bridge and the twin-paradox structural identification," GeoVac corpus (2026).

[32] J. Loutey, "Paper 55: Periods of GeoVac," GeoVac corpus (2026).